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Synergistic bulky substituent strategy in iminopyridyl nickel and palladium catalysts enables High-Molecular-Weight (Co)polymers and living ethylene polymerization

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ABSTRACT

High-molecular-weight polyethylene is notoriously difficult to achieve with conventional iminopyridyl Ni(II) and Pd(II) catalysts, primarily because the ligand's one-sided axial steric hindrance is insufficient to suppress chain-transfer reactions. Particularly, in contrast to the extensively explored α-diimine Ni(II) systems widely employed for olefin-polar monomer copolymerization, the iminopyridyl Ni(II) system has scarcely been reported for ethylene copolymerization with polar monomers. Herein, we introduce a new class of iminopyridyl Ni(II) and Pd(II) complexes incorporating both "half-sandwich-type" naphthyl moieties and 4-methylphenyl (aryl) substituents. By incorporating aryl substituents at both the 2- and 8-naphthyl positions to provide coordinated axial blocking, these complexes display a synergistic steric effect that significantly suppresses chain-transfer reactions, thereby enabling the synthesis of high-molecular-weight (co)polymers. Notably, the Ni(II) catalysts enable efficient ethylene–MU copolymerization under mild conditions, producing high-molecular-weight copolymers (up to 88,000 g·mol^{-1})—a rare achievement for conventional iminopyridyl Ni(II) systems. Meanwhile, the corresponding Pd(II) analogues exhibit living polymerization behavior during ethylene homopolymerization at temperatures above ambient. Overall, the strategic combination of half-sandwich-type naphthyl units with *ortho*-aryl substituents in these iminopyridyl catalysts significantly enhances their catalytic performance.

1. INTRODUCTION

Late-transition-metal catalysts—particularly Ni(II) and Pd(II) complexes with α-diimine or iminopyridyl-type ligands—have attracted widespread interest for synthesizing branched and functionalized polyolefins by virtue of their chain-walking capability and high tolerance to polar monomers[1–9]. Compared to the widely explored α-diimine Ni/Pd systems (Scheme 1, **I**), iminopyridyl analogues often suffer from only partial axial shielding due to their inherently one-sided steric design (Scheme 1, **II**), which usually leads to lower polymer molecular weights or even oligomers [10–21]. The initial study by Laine *et al.* demonstrated that iminopyridyl Ni(II) catalysts could yield low molecular weight ethylene oligomer (Scheme 1, **III**, M_n < 3500 g·mol^{-1}). [22,23] Despite several reported strategies to enhance steric bulk—for instance, introducing substituents at the ortho position of the aniline moiety, which significantly.

increases molecular weight in α–diimine Ni(II) and Pd(II) systems [24–30]—the resulting (co)polymers in iminopyridyl Ni(II) and Pd(II) systems often remain low in molecular weight. For example, while substituting bulkier dibenzhydryl or diarylmethyl groups for diisopropyl in α–diimine catalysts can yield ultrahigh molecular weight polyethylene [24–27](UHMWPE) (Scheme 1, **IV** vs. **V**), applying the same strategy in iminopyridyl catalysts results in very low molecular weight polyethylene [20,31,32] (Scheme 1, **VI**, M_n < 8300 g·mol^{-1}). Overall, despite their high catalytic activity in ethylene polymerization, these iminopyridyl catalysts typically yield only oligomers or low-molecular-weight polyethylene because they incorporate only a single *ortho*-disubstituted aryl imine moiety, in contrast to the symmetric α-diimine systems that provide dual-sided steric hindrance (Scheme 1, **I** vs.**II**).

In recent years, several successful strategies have emerged to overcome the low molecular weight limitation in iminopyridyl catalyst systems. The earliest significant development was achieved by introducing "half-sandwich" 8-arylnaphthyl substituents into iminopyridyl Ni(II) catalysts [33,34] (Scheme 1, **VII**). These catalysts offered

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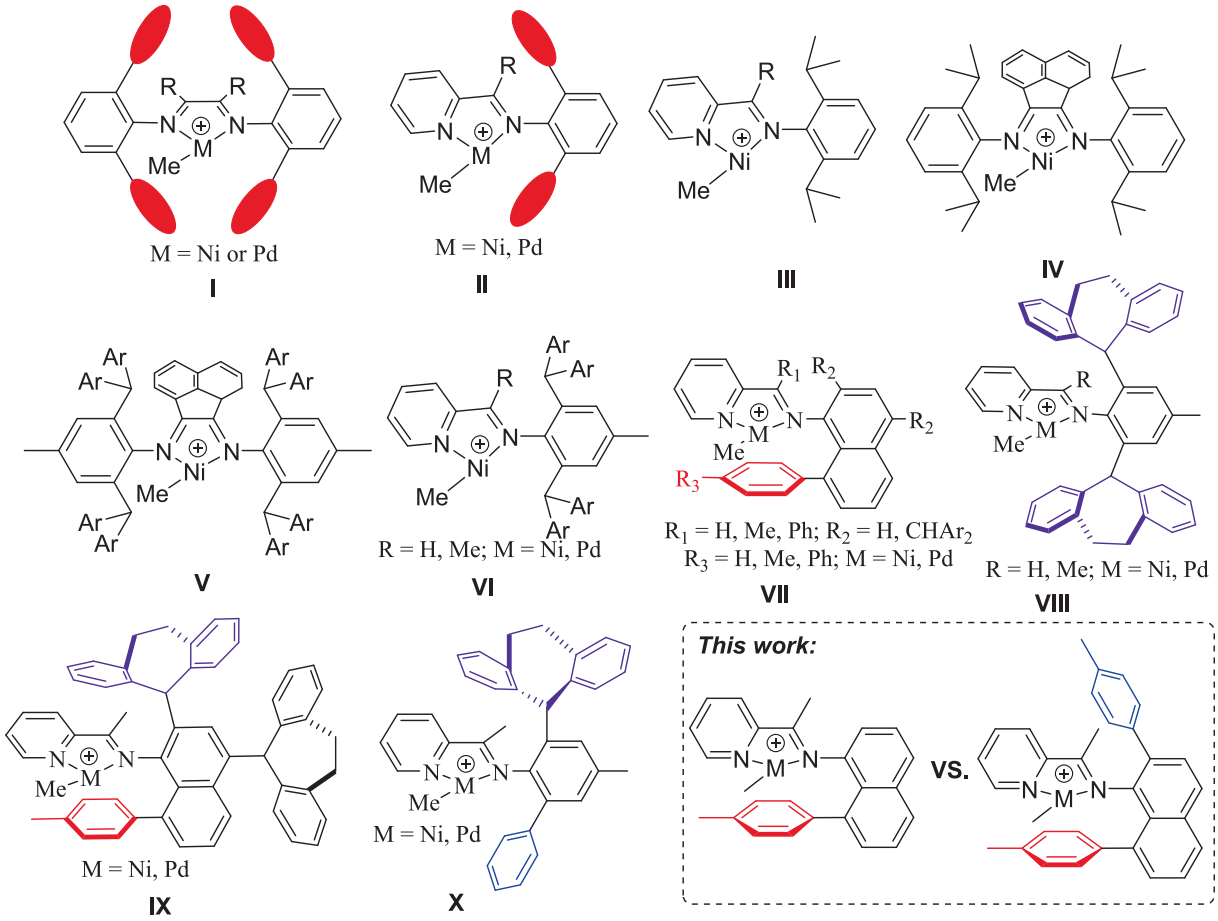
enhanced axial steric protection, which effectively suppressed chain-transfer processes and enabled the formation of polyethylene with significantly increased molecular weights. For example, Brookhart and co-workers demonstrated that a "half-sandwich" iminopyridyl catalyst incorporating an 8-arylnaphthylimino group (Scheme 1, **VII**, $R_1 = R_3 =$ Me, R_2 = H) significantly increased the molecular weight of polyethylene (M_n up to 26000 g·mol^{-1}) under their specific reaction conditions [34]. More recently, a rotation-restricted strategy employing dibenzosuberyl groups (Scheme 1, **VIII**) has been implemented to further suppress chain-transfer reactions, thereby enabling the synthesis of high-molecular-weight polyethylene or copolymers—with M_n values reaching up to 124000 g·mol^{-1}—in the iminopyridyl Ni(II) system [35,36]. Moreover, the synergistic effect achieved by incorporating both 8-arylnaphthyl and dibenzosuberyl substituents (Scheme 1, **IX**) leads to a more efficient suppression of chain-transfer reactions, enabling the production of high-molecular-weight polyethylene or copolymers even at elevated temperatures [36,37]. Additionally, iminopyridyl complexes bearing a combination of one ortho-dibenzosuberyl group and one *ortho*-phenyl substituent (Scheme 1, **X**) have also demonstrated superior chain-transfer suppression [38]. Notably, the polymers produced using this combination exhibit molecular weights (**X**, M_n = 185600 g·mol^{-1}) that exceed those obtained with either dibenzosuberyl alone (Scheme 1, **VIII**, M_n = 124000 g·mol^{-1}) or with dibenzosuberyl/8-arylnaphthyl combinations [38] (Scheme 1, **IX**, M_n = 56800 g·mol^{-1}), while catalysts featuring two phenyl substituents on the 2,6-position of N-aryl rings yield only low-molecular-weight oligomers [39,40]. These observations underscore the critical importance of the synergistic interplay between these substituents on the N-aryl ring. Consequently, we have adopted a strategy to synergistically combine 8-arylnaphthyl and phenyl (aryl) substituents to further enhance catalytic performance, outperforming catalysts that incorporate only 8-arylnaphthyl groups (Scheme 1, this work). This approach effectively suppresses chain transfer, thereby enabling the generation of high-molecular-weight (co)polymers and achieving living ethylene polymerization at conditions above ambient temperature.

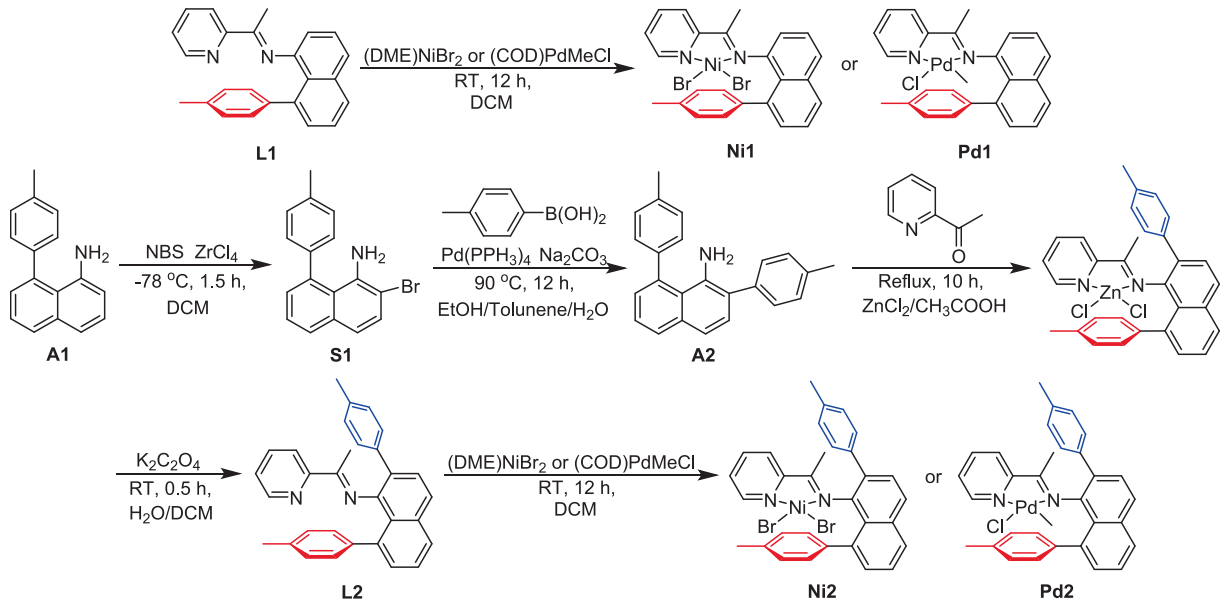
2. RESULTS AND DISCUSSION

2.1. Preparation and structural elucidation of ligands and complexes

Ligand **L1** was synthesized according to a previously published method [34]. 2,8-Di-p-tolylnaphthalen-1-amine (**A2**) was obtained by brominating 8-arylnaphthylamine, followed by a palladium-catalyzed coupling with arylboronic acids. Subsequently, the new iminopyridine ligand **L2** was synthesized using a template-type method [41,42] in which **A2** served as the starting material. This approach involves the formation of a zinc template complex, followed by removal of zinc chloride to yield **L2**. The ligand was then characterized by ^{1}H, ^{13}C NMR spectroscopy (Figs. S4 and S5) and mass spectrometry (Fig. S12). Building on the successful synthesis of ligands **L1** and **L2**, the corresponding nickel complexes were prepared in good yields (75–99 %) by reacting these ligands with an equimolar amount of (DME)$NiBr_2$ (DME = 1,2-dimethoxyethane) (Scheme 2. The identity and purity of the resulting nickel complexes were confirmed by using elemental analysis and mass spectrometry (Figs. S13). Likewise, **Pd1** and **Pd2** were synthesized in good yields (75–81 %) by reacting the ligands with (COD) PdMeCl (COD = 1,5-cyclooctadiene). These complexes were fully characterized using ^{1}H and ^{13}C NMR (Figs. S6-9), mass spectrometry



Scheme 1. Schematic overview comparing reported iminopyridyl Ni(II) and Pd(II) catalysts (**II**, **III**, **VI-X**), examples of ortho-substituent steric tuning in α-diimine systems (**IV** and **V**), and the strategy employed in our current work.



Scheme 2. Preparation of iminopyridyl ligands and the associated Ni(II)/Pd(II) complexes.

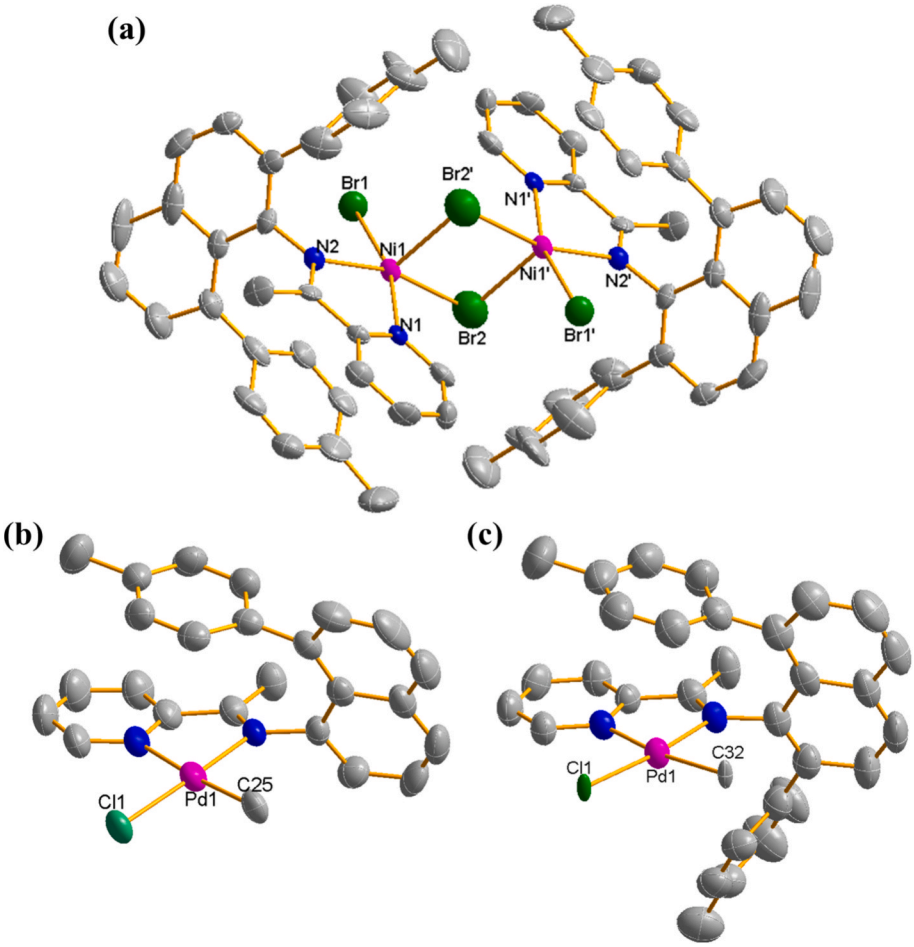


Fig. 1. The solid-state molecular structures of **Ni2** (a), **Pd1** (b), and **Pd2** (c) with 50 % probability level. H atoms have been omitted for clarity. Selected lengths (Å) and angles (deg): **Ni2** (a): Ni(1)-N(1) = 2.069(6), Ni(1)-N(2) = 2.107(7), Ni(1)-Br(1) = 2.4743(14), Ni(1)-Br(2) = 2.4965(18), N(1)-Ni(1)-N(2) = 77.0(3), Br(1)-Ni(1)-Br(2) = 91.97(5). **Pd1** (b): Pd(1)-N(2) = 2.043(4), Pd(1)-N(1) = 2.078(5), Pd(1)-C(25) = 2.196(17), Pd(1)-Cl(1) = 2.301(4), N(2)-Pd(1)-N(1) = 78.98(18), C(25)-Pd(1)-Cl(1) = 85.9(4). **Pd2** (c): Pd(1)-N(2) = 2.015(5), Pd(1)-N(1) = 2.031(5), Pd(1)-C(32) = 2.270(18), Pd(1)-Cl(1) = 2.293(3), N(2)-Pd(1)-N(1) = 79.6(2), C(32)-Pd(1)-Cl(1) = 91.5(4).

(Figs. S14-15), and elemental analysis. In the ^{1}H NMR spectra of the Pd complexes, the Pd–Me signal appears at 0.38 ppm for **Pd1** versus 0.21 ppm for **Pd2**. This difference may be attributed to the combined steric and electronic effects of the 4-methylphenyl substituent at the 2-position of the naphthyl ring. In particular, this substituent could increase the electron density at the palladium center while also enhancing the steric hindrance around it, both of which likely contribute to the observed shift differences. Previous studies indicate that iminopyridyl palladium complexes with dibenzhydryl substituents can exist as two geometric isomers, which arise from the methyl group being positioned either cis or trans to the bulkier side of the ligand [31]. In contrast, ^{1}H and ^{13}C NMR analyses of the palladium(II) complexes **Pd1** and **Pd2** reveal only a single isomer in solution. This observation is likely due to the substantial steric hindrance imposed by the bulky substituents, which favors one stable configuration and inhibits the formation of alternative isomers.

Single crystals of the complexes **Ni2**, **Pd1**, and **Pd2** were also obtained by layering their CH_2Cl_2 solution with diethyl ether or hexanes in a glovebox and were analyzed by X-ray diffraction (Fig. 1 and Table S4). The complex **Ni2** crystallized as a centrosymmetric dimer, in which each nickel center is coordinated to a pyridineimine ligand and two bridging halogen atoms (Fig. 1a), a behavior commonly observed in iminopyridyl nickel complexes [34]. The nickel center adopts a square-pyramidal coordination sphere. The aryl group (4-methylphenyl) at the 8-position is oriented nearly parallel to the five-membered chelate ring, positioning it in close proximity to the metal's axial sites and effectively blocking one axial coordination site. Moreover, the aryl groups at the 2- and 8-positions of the naphthyl ring form an angle of approximately 85° between their planes, indicating that the aryl group at the 2-position in complex **Ni2** is nearly perpendicular to the chelate ring (Fig. 2a). Unlike the nickel complexes, both **Pd1** and **Pd2** adopt square-planar mononuclear structures (Fig. 1b and 1c). In **Pd2**, similar to **Ni2**, one aryl group adopts a nearly parallel orientation to the chelate ring while the other is arranged in a different orientation, resulting in an interplanar angle of approximately 73° between the 2- and 8-aryl groups (Fig. 2b). As a result, the additional bulky substituents at the 2-position of the naphthyl ring in **Ni2** and **Pd2** primarily shield the nitrogen atom attached to the naphthalene ring, rather than directly blocking the axial coordination site, therefore suggesting that the overall structural arrangement still conforms to a half-sandwich configuration. However, compared with **Pd1**, **Pd2** provides more effective axial steric shielding, which is expected to further suppress chain-transfer reactions. Moreover, the introduction of the 2–position aryl group in **Pd2** may further stabilize the catalytic active center. Steric maps generated via the SambVca 2.1 software were used to quantitatively evaluate the spatial crowding surrounding the metal center (Fig. 3). By introducing 4-methylphenyl substituents at the 2-position of the naphthyl ring, the buried volume increased from 45.5 % (**Pd1**) to 52.9 % (**Pd2**) (Fig. 3a vs Fig. 3b). Under the same metal center conditions, the buried volume of **Pd2** is approximately 16 % higher than that of **Pd1**. For the nickel complex **Ni2**, the introduction of 4–methylphenyl substituents at the 2–position of the naphthyl ring resulted in a buried volume of 49.7 % (Fig. 3c), also indicating substantial steric congestion around the metal center. The steric shielding in **Pd2** and **Ni2** favors the suppression of chain-transfer process, thereby enhancing the molecular weight of the resulting polymer and potentially enabling living polymerization (see ethylene (co) polymerization results below).

2.2. Ni(II)-catalyzed ethylene homopolymerization and copolymerization

The in situ activation of Ni(II) complexes with a large excess of Et_2AlCl was carried out for ethylene polymerization, and the results are summarized in Table 1. Polymerization conditions were systematically varied, with temperatures screened from 30 to 70 °C under ethylene pressures of either 8 or 15 atm (Table 1, entries 1 − 8). Notably, these Ni(II) complexes demonstrated high catalytic activity, reaching up to 1.9 × 10^6 g/(mol Ni·h). Polymerization experiments conducted under identical temperatures revealed that **Ni1** exhibited higher activity than **Ni2**. This lower activity for **Ni2** can be attributed to the increased steric bulk of its 2,8-diarylnaphthyl moiety, which slows monomer trapping—a trend consistent with previous reports [20]. Notably, however, the enhanced steric hindrance at the ortho-position in **Ni2** resulted in a dramatic increase in polymer molecular weight. Under 8 atm ethylene pressure, the polyethylene produced by **Ni2** had an M_n 24–43 times higher than that from **Ni1** (M_n: 3000–6000 g·mol^{-1} for **Ni1** versus 72000–220000 g·mol^{-1} for **Ni2**). At 15 atm, the M_n for **Ni2**-derived polyethylene was 62 times higher than that for **Ni1** (M_n: 6000 vs. 373000 g·mol^{-1}). This significant improvement is primarily ascribed to the synergistic effect of the 8-arylnaphthyl and phenyl substituents in **Ni2**, which provide effective axial shielding and suppress chain-transfer reactions. Additionally, polyethylene from **Ni1** exhibited a slightly higher branching density (91–102 branches/1000C) compared to that from **Ni2** (84–96 branches/1000C). Consequently, the **Ni1**-derived polymer, with its higher branching and lower molecular weight, exhibited an oil-like morphology, whereas the **Ni2**-derived polymer was solid, with low melting points (−5.8 to 27.4 °C) and a broad melting

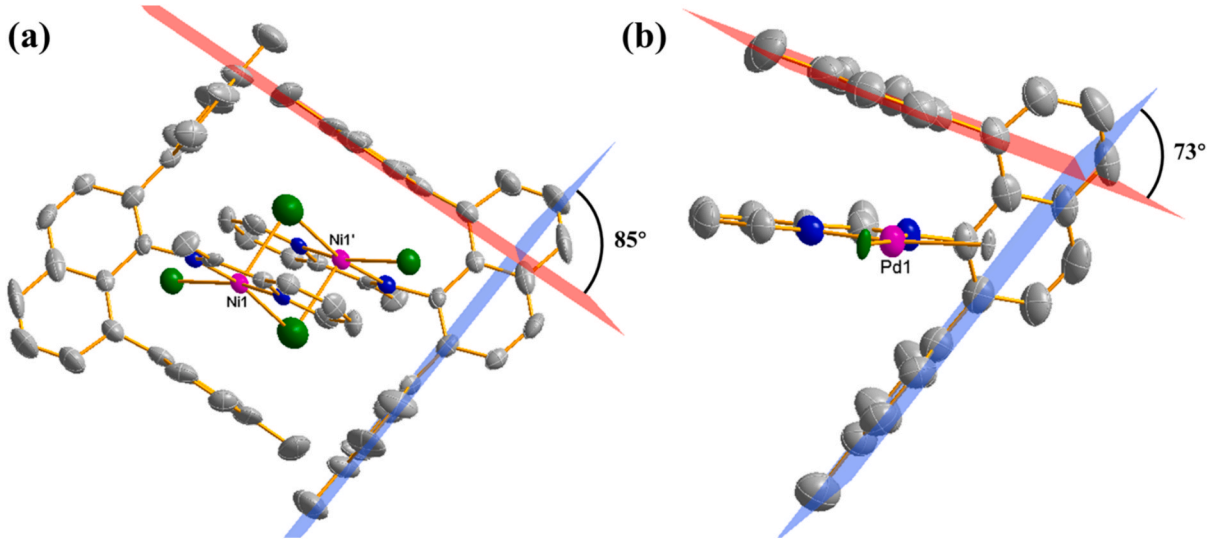


Fig. 2. The interplanar angle between the aryl groups at the 2- and 8-positions of the naphthyl ring in **Ni2** (a) and **Pd2** (b).

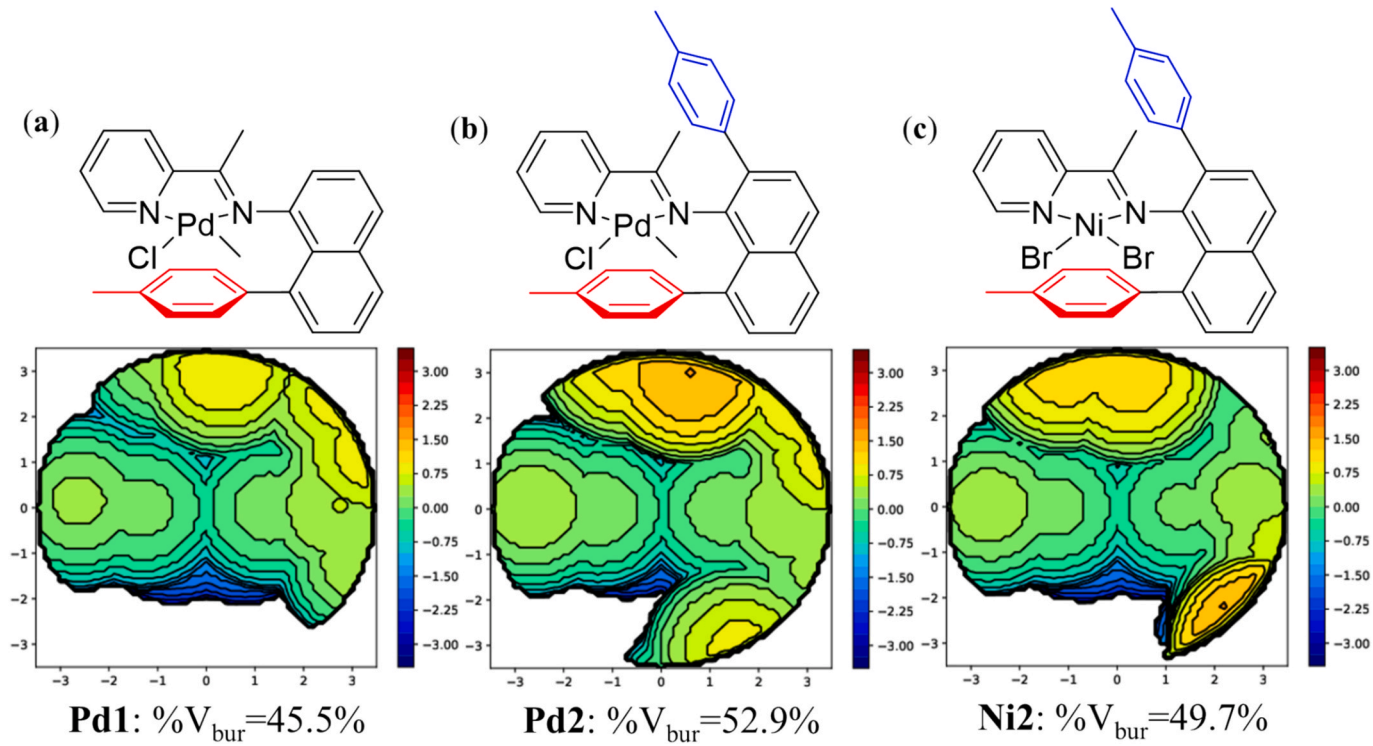


Fig. 3. Steric topographic maps of **Pd1** (a), **Pd2** (b) and **Ni2** (c) illustrating buried volume-based quantification of steric environments.

Table 1
Ethylene polymerization with **Ni1** and **Ni2**[a].

Ent.	Precat.	P/atm	T/°C	Yield/g	Act.[b]	M_n[c]	PDI[c]	B[d]	T_m[e]
1	**Ni1**	8	30	1.22	12.2	0.6	1.78	93	oil
2	**Ni1**	8	50	0.55	5.5	0.4	1.67	99	oil
3	**Ni1**	8	70	0.13	1.3	0.3	1.48	102	oil
4	**Ni1**	15	30	1.87	18.7	0.6	1.79	91	oil
5	**Ni2**	8	30	0.29	2.9	22.0	1.25	85	22.2
6	**Ni2**	8	50	0.28	2.8	17.0	1.69	89	2.9
7	**Ni2**	8	70	0.10	1.0	7.2	2.10	96	−5.8
8	**Ni2**	15	30	0.67	6.7	37.3	1.41	84	27.4

[a] Conditions: 2 μmol pre-catalyst, 1000 eq. $AlEt_2Cl$, 5 mL of CH_2Cl_2, 35 mL toluene, 8 atm, 30 min. [b]Activity (Act.) = 10^5 g/(mol h). [c]M_n is in unit of 10^4 g mol^{-1} and molecular weight was determined by GPC in trichlorobenzene at 40 °C using polystyrene standards. [d]B = branches per 1000 carbons, determined by ^{1}H NMR spectroscopy.Branching numbers for low molecular weight samples were corrected for end groups. [e]Determined by differential scanning calorimetry (DSC).

range. When the polymerization temperature increased from 30 °C to 70 °C, both the catalytic activity and the molecular weight of the polymers decreased, which is consistent with previous observations [33]. Although a slight increase in branching density was observed with rising temperature, the overall change was modest. In addition, under identical catalyst and temperature conditions, increasing ethylene pressure led to only slight changes in branching density (Table 1, entries 1 vs. 4 and entries 5 vs. 8). This suggests that, in this catalyst system, ethylene pressure has a limited influence on chain propagation over chain walking. The branching distribution of polyethylene obtained from **Ni1** and **Ni2** was further evaluated by ^{13}C NMR spectroscopy. According to literature assignments [43–49], polyethylene samples synthesized using **Ni1** exhibited multiple branch types including methyl (Me), long-chain branches (>C4, Lg), and other short-chain branches (ethyl, propyl, and butyl) (Table S1 and Fig. S39). In contrast, polyethylene derived from **Ni2** predominantly displayed only two types of branches—methyl (Me, 19.2 ppm) and long-chain branches (Lg, 13.3 ppm) (Table S1 and Fig. S40). This observation suggests that the incorporation of ethylene into secondary Ni(II)-alkyl species carrying ethyl, propyl, or butyl groups is likely inhibited due to steric constraints resulting from the enclosed steric environment around the Ni(II) center in **Ni2**. A similar phenomenon has been previously observed in α-diimine Ni(II) catalyst systems, where the branching distribution—characterized by the predominance of methyl and long-chain branches and the absence of intermediate short-chain branches—could be modulated by adjusting the steric bulk of *ortho*-substituents [50,51]. Thus, the current iminopyridyl Ni(II) catalyst system shows analogous steric control over branch-type distribution.

Achieving functionalized polymers via coordination–insertion copolymerization of olefins with polar comonomers has long posed a significant challenge in the field. Numerous research groups have demonstrated that Pd(II) or Ni(II) α-diimine catalysts can effectively incorporate polar monomers into olefin copolymerization, yielding high-molecular-weight, functionalized polyolefin [25,51–60]. Similarly, many studies have reported successful olefin copolymerizations using iminopyridyl Pd(II) catalysts [36,38,61–68], which exhibit superior tolerance for polar comonomers compared to the corresponding Ni(II) catalysts, although this often leads to a further reduction in polymer molecular weight. In contrast, to our knowledge, attempts using iminopyridyl Ni(II) systems for the synthesis of functionalized polyethylene have either not been explored or have yielded no catalytic activity. The copolymerization of ethylene with the inexpensive and widely used

monomer methyl acrylate (MA) was initially investigated using **Ni1** and **Ni2** at [MA] = 0.2 M. For **Ni1**, which features only an 8-arylnaphthyl substituent, the presence of MA nearly completely inhibited ethylene

polymerization (Table 2, entry 9 and Scheme 3). In contrast, **Ni2**, bearing a 2,8-diarylnaphthyl moiety, exhibited significant polymerization activity in the presence of MA—although no MA incorporation was observed (Table 2, entry 10 and Scheme 3). The increased steric bulk and enhanced stability of **Ni2** are likely responsible for this phenomenon. Due to the long spacer between its double bond and polar functional group, methyl undecenoate (MU) is more readily incorporated into the ethylene polymerization process—a strategy proven effective for certain early-transition-metal and nickel catalysts [54,69–72]. Accordingly, the copolymerization of ethylene with MU was investigated using both **Ni1** and **Ni2** catalysts (Table 2 entries 11–12 and Scheme 3). ^{1}H NMR analysis confirmed successful copolymer formation (Fig. 4), with both **Ni1** and **Ni2** exhibiting similar MU incorporation levels (approximately 0.6 % for **Ni1** and 0.5 % for **Ni2**) and nearly identical activities (approximately 6.4 × 10^4 g/(mol Ni·h)). However, consistent with the trend observed in ethylene homopolymerization, the copolymer produced by **Ni2** had a dramatically higher molecular weight—reaching up to 88000 g/mol, which is 88 times that of **Ni1** (1000 g/mol). Moreover, while the **Ni1**-derived copolymer exhibited an oil-like morphology, the **Ni2**-derived material was a low-melting solid (T_m: 2.7 °C) with a broad melting range. These differences can be attributed to the more effective steric shielding provided by the 2,8-diarylnaphthyl moiety in **Ni2**, which suppresses chain-transfer reactions more efficiently than the 8-arylnaphthyl substituent alone in **Ni1**.

2.3. Pd(II)-catalyzed ethylene homopolymerization and copolymerization

Pd1 and **Pd2** complexes were activated using an excess of sodium tetrakis(3,5-bis(trifluoromethyl)phenyl)borate (NaBArF) to catalyze ethylene polymerization and copolymerization with MA, performed at 30 °C under 8 or 15 atm of ethylene (Table 3, entries 13–16). Under these conditions, both **Pd1** and **Pd2** exhibited moderate activities (6.3–13.7 × 10^3 g/(mol Pd·h)) and produced highly branched polyethylene. Notably, at identical reaction conditions, **Pd2** showed higher activity than **Pd1**. Consistent with observations in Ni(II) systems, the ligand steric effect strongly influenced the polymer molecular weight: the polyethylene produced by **Pd2** was approximately one order of magnitude higher in M_n than that generated by **Pd1** (Table 3, entry 13 vs. entry 14, entry 15 vs. entry 16). Nevertheless, due to their high branching density, the polyethylene derived from **Pd2** remained as an amorphous oil. Furthermore, ^{13}C NMR analysis indicates that the polyethylene produced by both **Pd1** and **Pd2** exhibits a similar branching distribution, containing methyl, ethyl, n-propyl, n-butyl, and long-chain (>C4) branches in comparable proportions (Table S1, Figs. 41 and 42).

In ethylene–MA copolymerization, **Pd2** produced a copolymer with a molecular weight of approximately 13.9 × 10^3 g·mol^{-1}—about four times higher than that obtained with **Pd1** (3.5 × 10^3 g·mol^{-1}). This also highlights the critical role of synergistically combining 8-arylnaphthyl and 2-arylnaphthyl substituents in effectively suppressing chain transfer. Both **Pd1** and **Pd2** afforded copolymers with over 10 % MA incorporation (12.5 % for **Pd1** and 10.2 % for **Pd2**) at an MA concentration of 1.0 M, without a significant decrease in molecular weight compared to the corresponding homopolymers. The high efficiency in MA incorporation is most likely attributed to the open coordination environment around the palladium center, which arises from the relatively low axial hindrance of the pyridine moiety in the asymmetric iminopyridyl ligand compared to that in symmetric α-diimine systems (scheme 1, **I** vs. **II**). The calculated branching density of the copolymers (55–64 branches/1000C) is lower than that of the homopolymers. This is likely due to the higher MA incorporation ratio, with MA units predominantly located at the end of branches (Fig. S30), which is in agreement with previous reports on α–diimine Pd(II) catalysts [2,59,73,74]. Since the branching density is calculated based on terminal methyl branches, the terminal incorporation of MA (ester groups) results in an underestimation of the actual branching.

Notably, the polyethylene produced by **Pd2** exhibits a much narrower molecular weight distribution (M_w/M_n: 1.18–1.28) compared to that from **Pd1** (M_w/M_n: 1.49–1.52), suggesting a living polymerization pattern. Consequently, the living behavior of ethylene polymerization using **Pd2** was further investigated at 30 °C with NaBArF as the activator under 4 atm of ethylene, with the reaction quenched by methanol at the desired time (Table S2). Fig. 5 presents the evolution of M_n and M_w/M_n over polymerization time, together with the associated GPC traces. M_n exhibits a linear increase as the polymerization proceeds, and the M_w/M_n remains between 1.09 and 1.30 for up to 5 h—with values below 1.18 during 3 h—confirming a living polymerization process. Notably, after 5 h, polyethylene with an M_n of approximately 34400 g·mol^{-1} is formed. A further attempt at synthesizing a diblock copolymer, polyethylene-b-poly(1-hexene), was carried out using catalyst **Pd2** at 30 °C. After polymerizing ethylene for 2 h, the ethylene was removed, followed by the addition of 1-hexene and continued stirring for another 2 h. The final copolymer exhibited a molecular weight (M_n = 34.6 × 10^3 g·mol^{-1}) and a narrow dispersity (M_w/M_n = 1.29), which were higher than those of the polyethylene obtained at 2 h (M_n = 16.2 × 10^3 g·mol^{-1}, M_w/M_n = 1.14) and the poly(1-hexene) produced at 2 h (M_n = 26.6 × 10^3 g·mol^{-1}, M_w/M_n = 1.30) (Table S3, Fig. S1). These results further support the living nature of the **Pd2** catalyst in ethylene polymerization. Notably, living ethylene polymerization has been extensively reported with α-diimine nickel and palladium catalysts—particularly in palladium systems, although these generally require sub-ambient conditions [53,75–80]. To the best of our knowledge, this work represents the first example of an iminopyridyl palladium catalyst (**Pd2**) achieving living ethylene polymerization, particularly under conditions above room temperature. Obviously, these results are attributed to the synergistic steric hindrance provided by the catalyst's design, which effectively suppresses chain-transfer reactions and enables the living polymerization behavior.

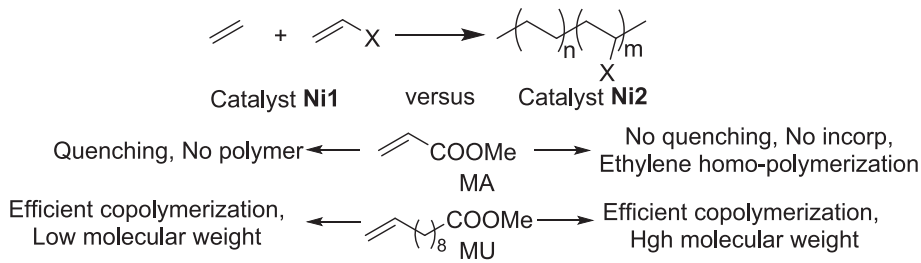
3. CONCLUSIONS

In summary, a novel class of 2,8-diarylnaphthyl iminopyridyl Ni(II) and Pd(II) complexes featuring both half-sandwich-type naphthyl and

Table 2
Ethylene–polar monomer copolymerization with **Ni1** and **Ni2**[a].

Ent.	Precat.	Monomer	Yield/g	Act.[b]	X[c] (mol %)	M_n[d]	PDI[d]	B[e]	T_m[f]
9	**Ni1**	MA	Trace	–	–	–	–	–	–
10	**Ni2**	MA	3.86	6.4	0	1.0	1.71	91	2.7
11	**Ni1**	MU	3.84	6.4	0.6	0.1	1.51	102	oil
12	**Ni2**	MU	3.86	6.4	0.5	8.8	1.87	86	2.7

[a] Conditions: 20 μmol pre-catalyst, polar monomer concentration: 0.2 M, 600 eq. $AlEt_2Cl$, 5 mL of CH_2Cl_2, total volume of toluene and monomer: 35 mL, 4 atm, 3 h, 30 °C. [b]Activity (Act.) = 10^4 g/(mol h). [c]Determined by ^{1}H NMR spectroscopy. [d]M_n is in unit of 10^4 g mol^{-1} and molecular weight was determined by GPC in trichlorobenzene at 40 °C using polystyrene standards. [e]B = branches per 1000 carbons, determined by ^{1}H NMR spectroscopy, Branching numbers for low molecular weight samples were corrected for end groups. [f]Determined by differential scanning calorimetry (DSC).



Scheme 3. Comparison of the ethylene-polar monomer copolymerization properties of **Ni1** and **Ni2**.

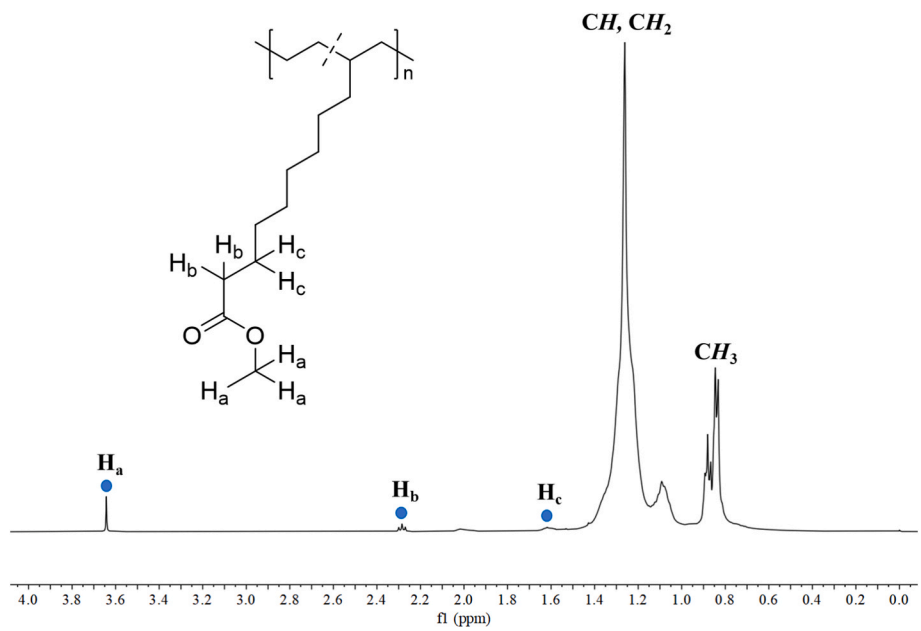


Fig. 4. ^{1}H NMR spectrum of ethylene-MU copolymer generated by **Ni2** (Table 2, entry 12).

Table 3
Ethylene polymerization[a] and ethylene–polar monomer copolymerization[b] with **Pd1** and **Pd2**.

Ent.	Precat.	Monomer	[M] (mol/L)	p/atm	T/h	Yield/g	Act.[c]	X[d] (mol %)	M_n[e]	PDI[e]	B[f]
13	**Pd1**	–	0	4	3	0.19	6.3	–	2.8	1.49	82
14	**Pd2**	–	0	4	3	0.33	11.0	–	25.6	1.18	106
15	**Pd1**	–	0	15	3	0.28	9.3	–	2.7	1.52	98
16	**Pd2**	–	0	15	3	0.41	13.7	–	26.4	1.28	92
17	**Pd1**	MA	1	4	8	0.18	1.1	12.5	3.5	1.33	55
18	**Pd2**	MA	1	4	8	0.27	1.7	10.2	13.9	1.24	64

[a] Conditions: 10 μmol pre-catalyst, 2 eq. NaBArF, 40 mL of CH_2Cl_2, 4 atm, 30 °C. [b]Conditions: 20 μmol pre-catalyst, 2 eq. NaBArF, 30 mL of CH_2Cl_2, 4 atm, 8 h, 30 °C. [c]Activity (Act.) = 10^3 g/(mol h). [d]Determined by 1H NMR spectroscopy. [e]M_n is in unit of 10^3 g mol^{-1} and molecular weight was determined by GPC in trichlorobenzene at 40 °C using polystyrene standards. [f]B = branches per 1000 carbons, determined by ^{1}H NMR spectroscopy. Branching numbers for low molecular weight samples were corrected for end groups.

ortho-aryl substituents was developed to address the limitations of conventional iminopyridyl catalysts. By incorporating aryl substituents at both the 2- and 8-positions to provide coordinated axial blocking, these complexes exhibit a synergistic steric effect that significantly suppresses chain-transfer reactions, thereby enabling the synthesis of high-molecular-weight polymers, whether in ethylene homopolymerization or copolymerization with polar monomers. Specifically, the Ni(II) catalyst (**Ni2**) produces polyethylene with molecular weights up to 62 times higher than that obtained with **Ni1**, while the Pd(II) catalyst (**Pd2**) not only yields polymers with substantially increased molecular weight but also demonstrates living polymerization behavior at conditions above room temperature. In addition, it was observed that the iminopyridyl nickel catalysts employed in this study successfully catalyze the copolymerization of ethylene with the polar monomer MU, yielding copolymers with high molecular weight (up to 88000 g·mol^{-1}) and the observed T_m. This contrasts with previous reports, in which such copolymerization using iminopyridyl nickel catalysts was either unexplored or found to be inactive.

4. Experimental section

General considerations and the synthetic procedure for intermediate compounds (anilines **S1** and **A2**) are shown in Supplementary Materials. **L1** and **Ni1** were synthesized using the reported procedures [34,81].

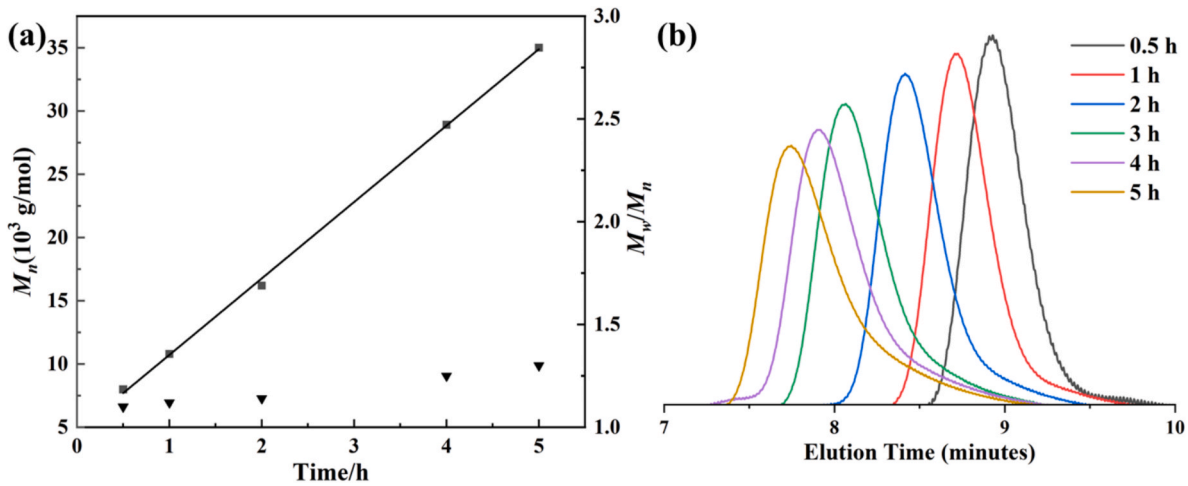
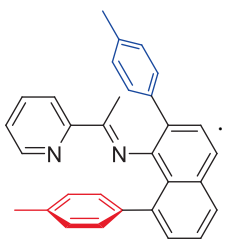


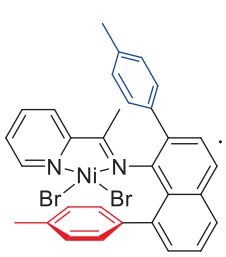
Fig. 5. (a) Plot of M_n (■) and M_w/M_n (▼) as a function of polymerization time at 30 °C under 4 atm of ethylene; (b) catalyst **Pd2** GPC traces at different times.

4.1. Synthesis of pro-ligands



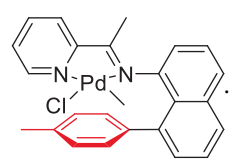
L2: Synthesis of the target compound was carried out in a Schlenk flask by sequentially adding $ZnCl_2$ (0.6 g, 4.65 mmol), 2-acetylpyridine (0.56 g, 4.65 mmol), 2,8-di-p-tolylnaphthalen-1-amine (1.04 g, 3.10 mmol) and glacial acetic acid (20 mL). The reaction mixture was maintained under reflux conditions for 12 h. Upon cooling to ambient temperature, the resulting yellow precipitate was collected by filtration and subsequently washed with hexanes (3 × 10 mL). The crude product was then dissolved in dichloromethane (100 mL) in a round-bottom flask. To this solution was added an aqueous solution of potassium oxalate monohydrate (6.0 g in 20 mL water) under continuous stirring. After 30 min of vigorous stirring, the organic layer was separated and subjected to multiple aqueous washes (3 × 10 mL H_2O), followed by drying over anhydrous magnesium sulfate. Purification was achieved through column chromatography using a petroleum ether/ethyl acetate (20:1) solvent system, yielding the desired product as a yellow solid (0.3 g, 23 %). ^{1}H NMR (500 MHz, Chloroform-*d*) δ 8.39 (d, *J* = 4.8 Hz, 1H), 7.86 (d, *J* = 8.1 Hz, 1H), 7.72 (d, *J* = 8.4 Hz, 1H), 7.67 (d, *J* = 7.9 Hz, 1H), 7.56 – 7.48 (m, 2H), 7.45 (t, *J* = 7.6 Hz, 1H), 7.28 (d, *J* = 7.9 Hz, 2H), 7.20 (d, *J* = 7.0 Hz, 1H), 7.17 (d, *J* = 7.3 Hz, 2H), 6.99 (d, *J* = 7.8 Hz, 2H), 6.93 (d, *J* = 7.7 Hz, 1H), 6.85 (d, *J* = 7.7 Hz, 1H), 6.61 (d, *J* = 7.6 Hz, 1H), 2.22 (s, 3H, Ar-C*H*$_3$), 1.87 (s, 3H, Ar-C*H*$_3$), 1.66 (s, 3H, Ar-C(C*H*$_3$)=N). ^{13}C NMR (126 MHz, Chloroform-*d*) δ 165.18 (*C*=N), 155.34, 147.59, 145.64, 141.98, 140.17, 138.05, 135.81, 135.11, 135.01, 134.61, 130.42, 129.80, 129.70, 129.50, 128.78, 128.63, 128.03, 127.93, 127.58, 127.05, 126.10, 125.03, 124.25, 123.48, 121.83, 21.15 (Ar-*C*H$_3$), 20.93 (Ar-*C*H$_3$), 17.47 (Ar-C(*C*H$_3$)=N). ESI-MS (*m/z*): calcd for $C_{31}H_{26}N_2Na$, 449.1994; found, 449.1978 $[M + Na]^+$.

4.2. Synthesis of Ni(II) complexes

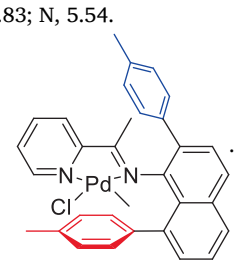


Ni2: Under inert atmosphere conditions within a glovebox, (E)-N-(2,8-di-p-tolylnaphthalen-1-yl)-1-(pyridin-2-yl)ethan-1-imine (0.20 g, 0.47 mmol) and (DME)$NiBr_2$ (0.15 g, 0.47 mmol) were stirred in dry dichloromethane (2 mL) for 24 h at room temperature. Following the reaction period, hexane (20 mL) was introduced to the homogeneous solution to induce precipitation. It was filtered to obtain a yellow solid, which was washed with hexanes (3 × 10 mL) and dried under a vacuum. The yellow solid was obtained (0.30 g, 99 %). ESI-MS (*m/z*): calcd for $C_{31}H_{26}BrN_2Ni$, 563.0633; found, 563.0642 $[M-Br]^+$. Anal. Calcd for $C_{31}H_{26}Br_2N_2Ni$: C, 57.72; H, 4.06; N, 4.34. Found: C, 57.52; H, 3.98; N, 4.39.

4.3. Synthesis of Pd(II) complexes



Pd1: Under inert atmosphere conditions within a glovebox, (E)-N-(2,8-di-p-tolylnaphthalen-1-yl)-1-(pyridin-2-yl)ethan-1-imine (0.21 g, 0.50 mmol) and (COD)PdMeCl (0.13 g, 0.50 mmol) were stirred in dry dichloromethane (2 mL) for 24 h at room temperature. Following the reaction period, hexane (20 mL) was introduced to the homogeneous solution to induce precipitation. It was filtered to obtain a yellow solid, which was washed with hexanes (3 × 10 mL) and dried under a vacuum. The yellow solid was obtained (0.22 g, 75 %). ^{1}H NMR (500 MHz, Chloroform-*d*) δ 9.07 (d, *J* = 7.1 Hz, 1H), 7.95 (d, *J* = 8.3 Hz, 1H), 7.94 – 7.83 (m, 4H), 7.58 – 7.50 (m, 5H), 7.31 (d, *J* = 7.0 Hz, 2H), 6.97 (d, *J* = 7.3 Hz, 1H), 6.89 (d, *J* = 9.7 Hz, 1H), 6.51 (d, *J* = 7.8 Hz, 1H), 1.97 (s, 3H, Ar-C*H*$_3$), 1.84 (s, 3H, Ar-C(C*H*$_3$)=N), 0.38 (s, 3H, Pd-C*H*$_3$). ^{13}C NMR (126 MHz, Chloroform-*d*) δ 172.29 (*C*=N), 152.89, 148.60, 143.47, 139.35, 138.15, 137.67, 136.27, 135.01, 131.87, 130.76, 130.17, 128.92, 128.14, 127.56, 127.10, 126.89, 125.69, 125.01, 124.05, 123.89, 120.63, 20.78 (Ar-*C*H$_3$), 19.08 (Ar-C(*C*H$_3$)=N), 0.15 (Pd-*C*H$_3$). ESI-MS (*m/z*): calcd for $C_{25}H_{23}N_2Pd$, 457.0896; found, 457.0925 $[M-Cl]^+$. Anal. Calcd for: C, 60.87; H, 4.70; N, 5.68. Found: C, 60.63; H, 4.83; N, 5.54.



Pd2: The palladium complex **Pd2** was synthesized following an identical protocol to that used for **Pd1** preparation, yielding the target compound as a crystalline yellow solid (0.20 g, 81 %). ^{1}H NMR (500 MHz, Chloroform-*d*) δ 9.00 (d, *J* = 4.3 Hz, 1H), 8.25 (d, *J* = 7.8 Hz, 1H), 7.91 (dd, *J* = 15.1, 7.6 Hz, 2H), 7.80 (t, *J* = 7.8 Hz, 1H), 7.69 (d, *J* = 7.9 Hz, 2H), 7.54 – 7.46 (m, 3H), 7.33 (d, *J* = 6.4 Hz, 2H), 7.12 (d, *J* = 7.7 Hz, 1H), 7.08 (d, *J* = 7.8 Hz, 2H), 6.67 (d, *J* = 7.6 Hz, 1H), 6.39 (d, *J* = 7.8 Hz, 1H), 2.25 (s, 3H, Ar-C*H*$_3$), 1.86 (s, 3H, Ar-C*H*$_3$), 1.76 (s, 3H, Ar-C(C*H*$_3$)=N), 0.21 (s, 3H, Pd-C*H*$_3$). ^{13}C NMR (126 MHz, Chloroform-*d*) δ 172.12 (*C*=N), 152.81, 148.70, 140.92, 140.64, 137.55, 137.05, 136.45, 135.93, 134.15, 131.50, 131.46, 130.80, 129.60, 129.51, 129.26, 128.68, 128.50, 127.81, 127.53, 126.62, 125.77, 124.29, 123.88, 21.28 (Ar-*C*H$_3$), 20.90 (Ar-*C*H$_3$), 19.87 (Ar-C(*C*H$_3$)=N), −1.80 (Pd-C*H*$_3$). ESI-MS (*m/z*): calcd for $C_{32}H_{29}N_2Pd$, 547.1366; found, 547.1387 $[M-Cl]^+$. Anal. Calcd for $C_{32}H_{29}ClN_2Pd$: C, 65.87; H, 5.01; N, 4.80. Found: C, 65.62; H, 4.96; N, 4.84.

4.4. Standard protocol for ethylene polymerization catalyzed by Nickel (II) complexes

Polymerization reactions were carried out in a mechanically agitated 100 mL high-pressure Parr reactor system with integrated gas delivery capabilities. Prior to use, the reaction vessel underwent thorough drying at 80°C under reduced pressure until mass stabilization was observed, ensuring complete moisture removal. The reaction system was initially charged with ethylene gas to 1.3–2.0 atm pressure. Maintaining this ethylene environment, 35 mL of anhydrous toluene and 2 mL of diethylaluminum chloride ($AlEt_2Cl$, 1000 equiv.) were sequentially introduced. After temperature equilibration at predetermined setpoints (30, 50, and 70 °C), the polymerization was initiated by rapid injection of a dichloromethane solution (5.0 mL) containing the Ni(II) precatalyst (2 μmol, 1 equiv.) following a 5-minute thermal stabilization period. The system pressure was rapidly elevated to 8 atm with continuous ethylene feed while maintaining precise temperature control. Vigorous stirring was maintained throughout the polymerization process. After 0.5 h of reaction time, the system was carefully vented to atmospheric pressure. The polymerization was quenched by addition of methanol containing 5 % hydrochloric acid (v/v), leading to immediate product precipitation. The resulting polymer was collected by vacuum filtration, followed by extensive washing with methanol. Subsequent drying under reduced pressure at ambient temperature (24 h) yielded the final product with constant mass.

4.5. Standard protocol for Ethylene/Polar monomer copolymerization catalyzed by Nickel(II) complexes

Polymerization reactions were carried out in a mechanically agitated 100 mL high-pressure Parr reactor system with integrated gas delivery capabilities. Prior to use, the reaction vessel underwent thorough drying at 80 °C under reduced pressure until mass stabilization was observed, ensuring complete moisture removal. The reaction system was initially charged with ethylene gas to 1.3–2.0 atm pressure. Maintaining this ethylene environment, anhydrous toluene (exact volume specified in Table 3 notes), polar monomer and diethylaluminum chloride ($AlEt_2Cl$, 600 equiv.) were sequentially introduced. After temperature equilibration at predetermined setpoints (30 °C), the polymerization was initiated by rapid injection of a dichloromethane solution (5.0 mL) containing the Ni(II) precatalyst (20 μmol, 1 equiv.) following a 5-minute thermal stabilization period. The system pressure was rapidly elevated to 4 atm with continuous ethylene feed while maintaining precise temperature control. Vigorous stirring was maintained throughout the polymerization process. After 3 h of reaction time, the system was carefully vented to atmospheric pressure. The polymerization was quenched by addition of methanol containing 5 % hydrochloric acid (v/v), leading to immediate product precipitation. The resulting polymer was collected by vacuum filtration, followed by extensive washing with methanol. Subsequent drying under reduced pressure at ambient temperature (24 h) yielded the final product with constant mass.

4.6. Standard protocol for ethylene polymerization catalyzed by palladium (II) complexes

Polymerization reactions were carried out in a mechanically agitated 100 mL high-pressure Parr reactor system with integrated gas delivery capabilities. Prior to use, the reaction vessel underwent thorough drying at 80°C under reduced pressure until mass stabilization was observed, ensuring complete moisture removal. The reaction system was initially charged with ethylene gas to 1.5 atm pressure. The polymerization system was established under continuous ethylene flow by first charging the reactor with anhydrous dichloromethane (35 mL) and sodium tetrakis[3,5-bis(trifluoromethyl)phenyl]borate (NaBArF, 2 equiv.). After thermal equilibration to the target temperature of 30 °C (±0.5 °C) using a circulating bath, the reaction mixture was allowed to stabilize for precisely 5 min under constant agitation (300 rpm). The catalytic cycle was then initiated via rapid syringe injection of a dichloromethane solution (5.0 mL) containing the palladium(II) precatalyst (10 μmol, 1 equiv.). The system pressure was rapidly elevated to 4 atm with continuous ethylene feed while maintaining precise temperature control. Vigorous stirring was maintained throughout the polymerization process. After 3 h of reaction time, the system was carefully vented to atmospheric pressure. The polymerization was quenched by methanol, leading to immediate product precipitation. The resulting polymer was collected by vacuum filtration, followed by extensive washing with methanol. Subsequent drying under reduced pressure at ambient temperature (24 h) yielded the final product with constant mass. The isolation procedure commenced with solvent removal through rotary evaporation at reduced pressure (30 °C). The viscous crude product was subsequently subjected to chromatographic purification using a silica gel stationary phase (200–400 mesh) with petroleum ether (60–90 °C fraction) as the mobile phase in a flash chromatography system.

4.7. Standard protocol for Ethylene/Polar monomer copolymerization catalyzed by Palladium(II) complexes

Polymerization reactions were carried out in a mechanically agitated 100 mL high-pressure Parr reactor system with integrated gas delivery capabilities. Prior to use, the reaction vessel underwent thorough drying at 80°C under reduced pressure until mass stabilization was observed, ensuring complete moisture removal. The reaction system was initially charged with ethylene gas to 1.5 atm pressure. The polymerization system was established under continuous ethylene flow by first charging the reactor with anhydrous dichloromethane (exact volume specified in Table 3 notes), polar monomer and sodium tetrakis[3,5-bis(trifluoromethyl)phenyl]borate (NaBArF, 2 equiv.). After thermal equilibration to the target temperature of 30 °C (±0.5 °C) using a circulating bath, the reaction mixture was allowed to stabilize for precisely 5 min under constant agitation (300 rpm). The catalytic cycle was then initiated via rapid syringe injection of a dichloromethane solution (5.0 mL) containing the palladium(II) precatalyst (20 μmol, 1 equiv.). The system pressure was rapidly elevated to 4 atm with continuous ethylene feed while maintaining precise temperature control. Vigorous stirring was maintained throughout the polymerization process. Following an 8-hour polymerization period, the system was carefully vented to atmospheric pressure. The reaction was immediately quenched by addition of anhydrous methanol (50 mL), inducing rapid polymer precipitation. Final drying was performed under high vacuum (<0.1 mbar) at ambient temperature (23 ± 2°C) for 24 h, with periodic mass measurements confirming constant weight attainment (±0.5 % variation over 2 h).

CRediT authorship contribution statement

Wei Zhao: Writing – original draft, Investigation, Formal analysis, Data curation. **Ruixin Qin:** Investigation, Data curation. **Lihua Guo:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2025.114146.

Data availability

Data will be made available on request.

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